CS & IT ENGINEERING

Operating System

Memory Management

Lecture - 1

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Recap of Previous Lecture

Topic Deadlock Avoidance

Topic Banker’s Algorithm

Topic Deadlock Detection

Topics to be Covered

Topic Memory Management

Topic Memory Management Technique

Topic Contiguous Memory Management Technique

Topic : Banker's Algorithm

H.W. Questn

Process	Allocation	Max	Available	Need
	A B C	A B C	A B C	A B C
P_0	0 1 0	7 5 3	~~3 3 2~~ ~~2 3 0~~	7 4 3
P_1	~~2 0 0~~ 3 0 2	3 2 2	2 2 0 P1 5 2 2	0 2 0 ~~1 2 2~~
P_2	3 0 2	9 0 2	P3 7 4 3	6 0 0
P_3	2 2 1 ~~2 1 1~~	2 2 2		0 0 1 ~~0 1 1~~
P_4	0 0 2	4 3 3		4 3 1

safe state

[NAT]

#Q. What will happen if process P1 requests one additional instance of resource type A and two instances of resource type C ? $Req_1 \langle 1, 0, 2 \rangle$

What will happen if process P3 requests one additional instance of resource type B?

$Req_3 \langle 0, 1, 0 \rangle$

both requests are granted

Topic : Detection-Algorithm Usage

When should the deadlock detection be done? Frequently, or infrequently?

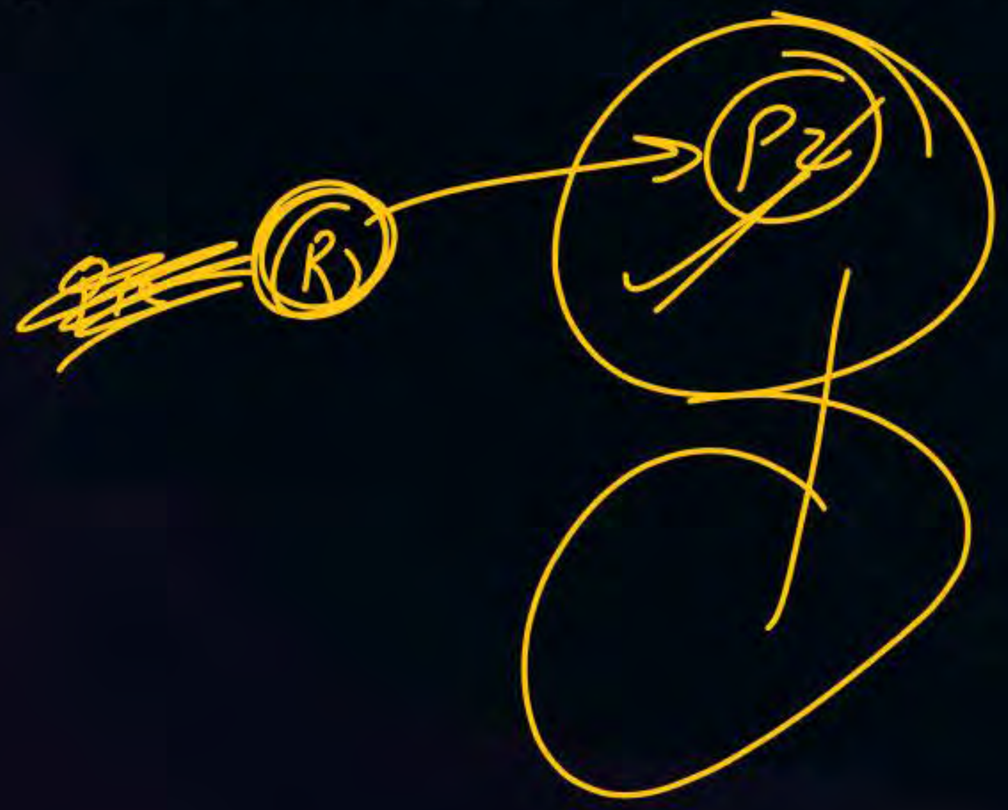
Topic : Detection-Algorithm Usage

1. Do deadlock detection after every resource allocation
2. Do deadlock detection only when there is some clue

Topic : Recovery From Deadlock

There are three basic approaches to recovery from deadlock:

1. Inform the system operator and allow him/her to take manual intervention
2. Terminate one or more processes involved in the deadlock
3. Preempt resources.



Topic : Process Termination

Terminate all processes involved in the deadlock

Terminate processes one by one until the deadlock is broken

Topic : Process Termination

Many factors that can go into deciding which processes to terminate next:

1. Process priorities.
2. How long the process has been running, and how close it is to finishing.
3. How many and what type of resources is the process holding
4. How many more resources does the process need to complete
5. How many processes will need to be terminated
6. Whether the process is interactive or batch

Topic : Resource Preemption

Important issues to be addressed when preempting resources to relieve deadlock:

1. Selecting a victim
2. Rollback
3. Starvation

[NAT]

Ans = $\underline{\underline{10}}$

#Q. Consider a system with 3 processes A, B and C. All 3 processes require 4 resources each to execute. The minimum number of resources the system should have such that deadlock can never occur?

	max	Allocated
A	4	3
B	4	3
C	4	3
		9

Available

1

for max 9 resources still deadlock may occur

for no deadlock

$9 + 1 = 10$

[NAT]

#Q. Consider a system with 4 processes A, B, C and D. All 4 processes require 6 resources each to execute. The maximum number of resources the system should have such that deadlock may occur?

	max	allocate
A	6	5
B	6	5
C	6	5
D	6	5
		20

Ans = 20

for deadlock not to occur ever

$= 20 + 1 = 21$

Ques) 5 Processes

	max	allocat^n
P_1	4	3
P_2	3	2
P_3	5	4
P_4	2	1
P_5	4	3

min no. of resources required so that deadlock can never occur?

Ans $= 3 + 2 + 4 + 1 + 3 + 1$

$= \underline{\underline{14}}$ Ans.

n no. of processes P_1 to P_n

max of each process $\Rightarrow$ max_1 to max_n

min. no. of resources needed such that deadlock never occurs

$$= \sum_{i=1}^{n} (max_i - 1) + 1$$

$$= \sum_{i=1}^{n} (max_i) - n + 1$$

[NAT]

#Q. Consider a system with 3 processes that share 4 instances of the same resource type. Each process can request a maximum of K instances. Resource instances can be requested and released only one at a time. The largest value of K that will always avoid deadlock is 2.

	max	allocatn
P1	k	k−1
P2	k	k−1
P3	k	k−1

$$3(k-1) + 1 \le 4$$

$$3k - 3 \le 3$$

$$3k \le 6$$

$$k \le 2$$

$$\boxed{k_{max} = 2}$$

Ques) n processes ⇒ each require 3 instances to execute

total resource ⇒ 15

max value of n such that no deadlock. ⇒ $\boxed{\boxed{\text{Ans} = 7}}$

Sol^n

$$n * (3-1) + 1 \leq 15$$

$$n * 2 \leq 14$$

$$n \leq 7$$

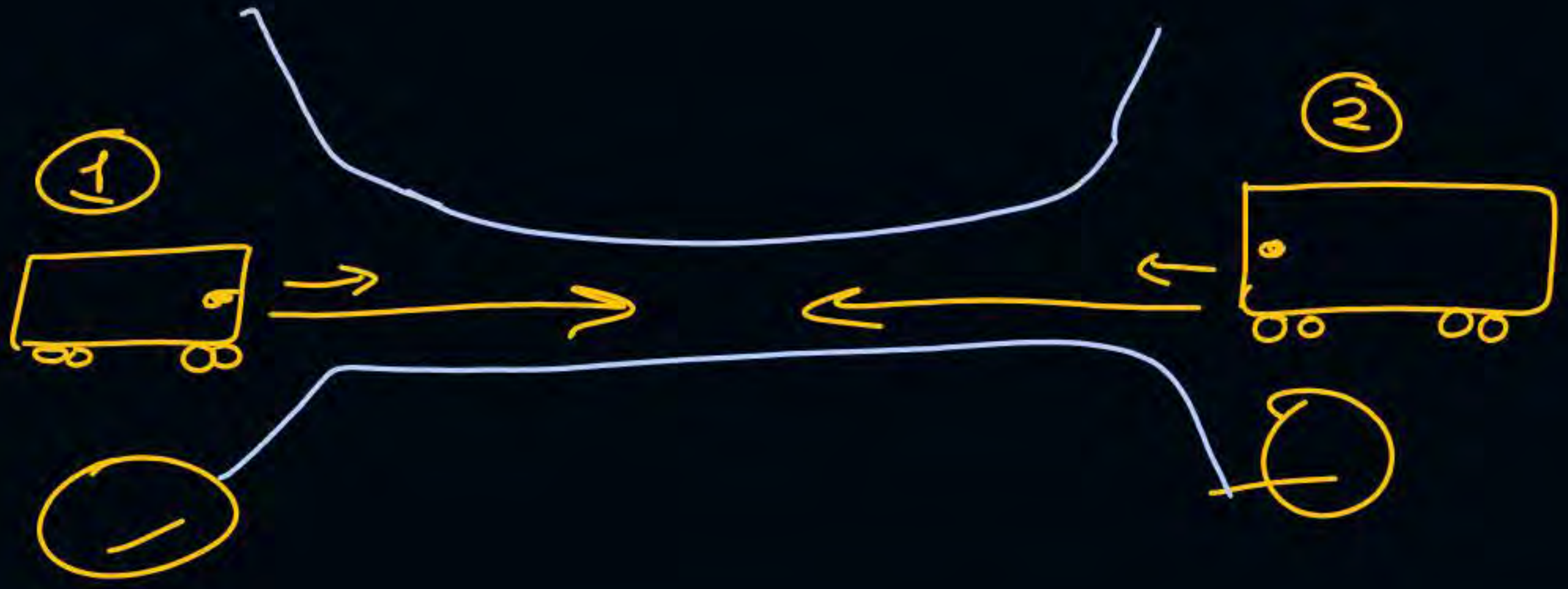
$$\boxed{n_{max} = 7}$$

Topic : Types of Locks

1. Spinlock ⇒ Busy waiting ex:- while (condition);
2. Livelock
3. Deadlock ⇒ in these processes can never execute further.
4. ~~Semaphores~~
5. Reentrant Locks

a process or thread can take same lock again on same item

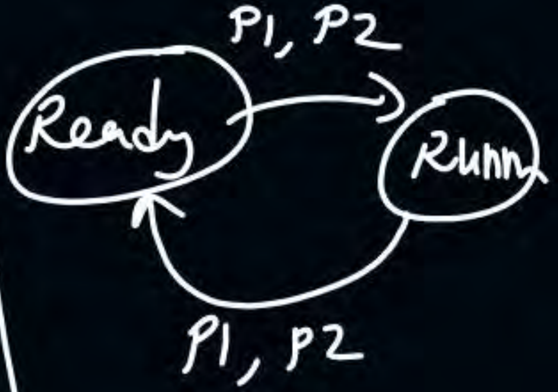
mostly refered as $\boxed{\text{deadlock}}$



P1	P2
wait(S1)	
	wait(S2)
wait(S2)	
	wait(S1)

$S1 = \cancel{1}\,0$

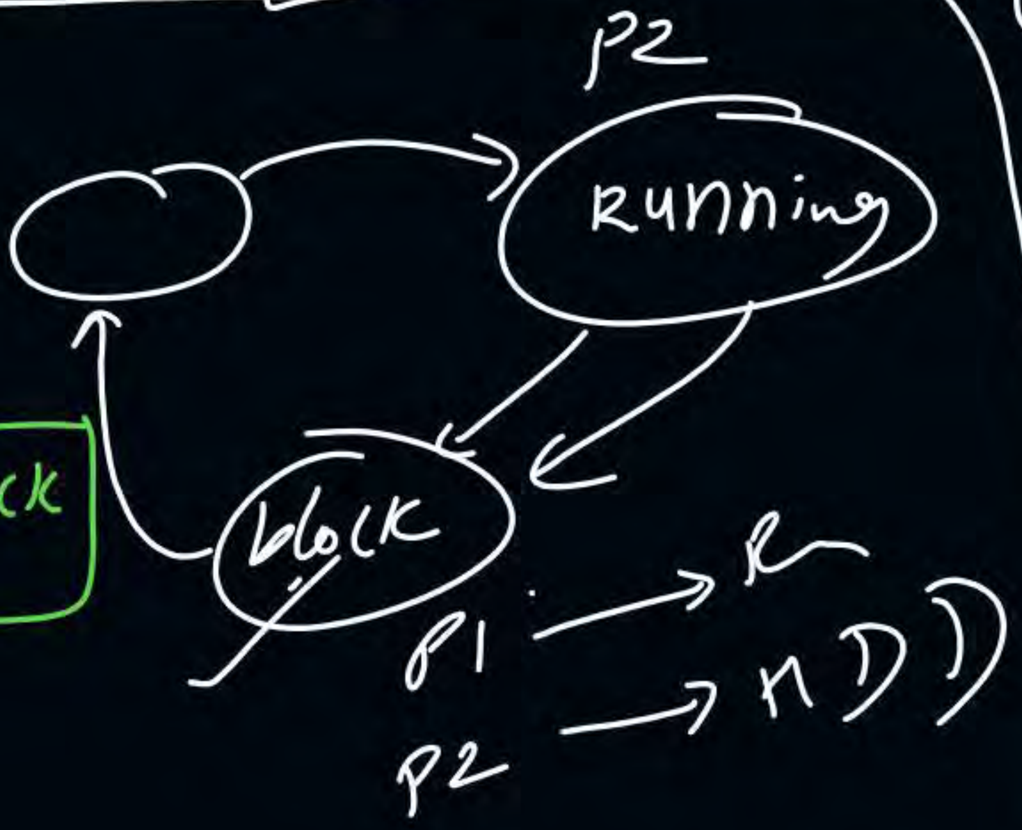
$S2 = \cancel{1}\,0$



	Hold	waiting
P1	HDD	Printer
P2	Printer	HDD

P1 and P2 ⇒ in blocked state

Deadlock



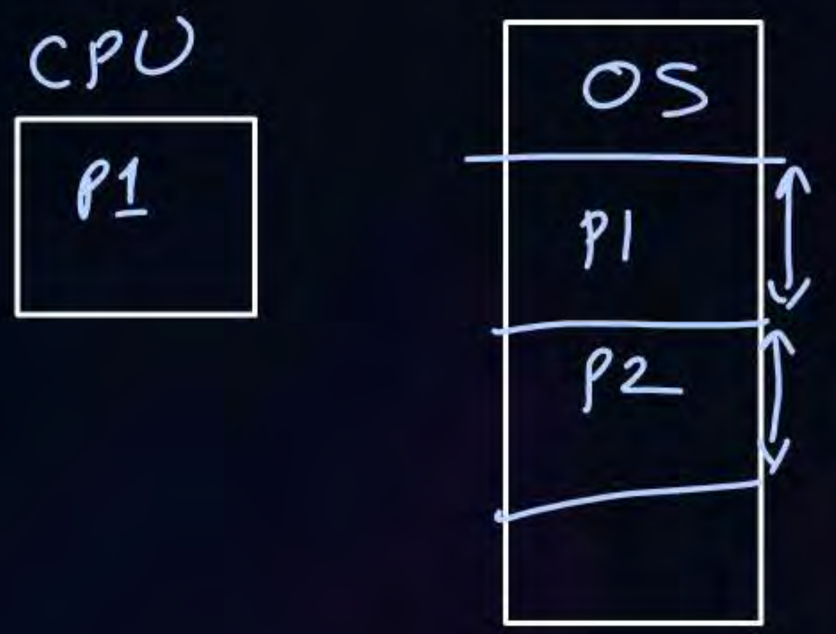
Topic : Memory Management

1. Module of OS

main memory (Primary mem.)

Topic : Functions of Memory Management

1. Memory allocation to new arriving process
2. Memory deallocation for completed process
3. Memory protection
   ↳ A process while executing in CPU can access only that part of memory which has been allocated to it.



Topic : Goals of Memory Management

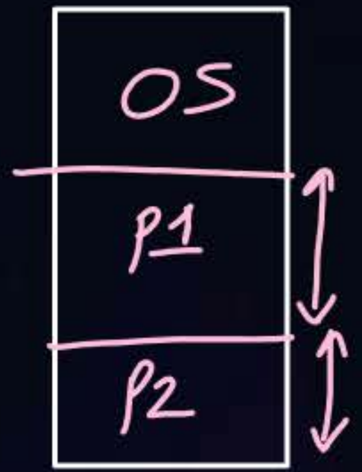
1. Maximum Utilization of space ⇒ min. wastage of space
2. Ability to run larger programs with limited space ⇒ through virtual memory

Topic : Memory Management Techniques

↳ How a new process allocated in memory

Contiguous

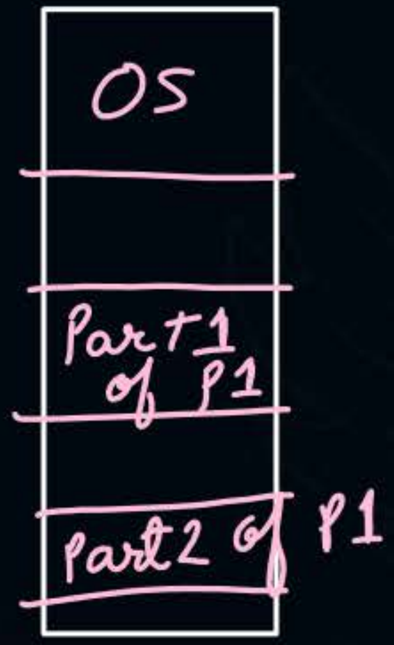
Entire process should be stored in memory on consecutive locations.



Types
- Fixed partition
- Variable partition

Non-Contiguous

A process can be stored on not consecutive mem. locations



Types
- Paging ☆
- Segmentation

Topic : Contiguous Memory Management

- Entire process should be stored on consecutive memory locations

Types

Fixed partition
contiguous m.m. technique

Variable partition
contiguous m.m. technique

2 mins Summary

Topic Memory Management

Topic Memory Management Technique

Topic Contiguous Memory Management Technique

Happy Learning

THANK - YOU